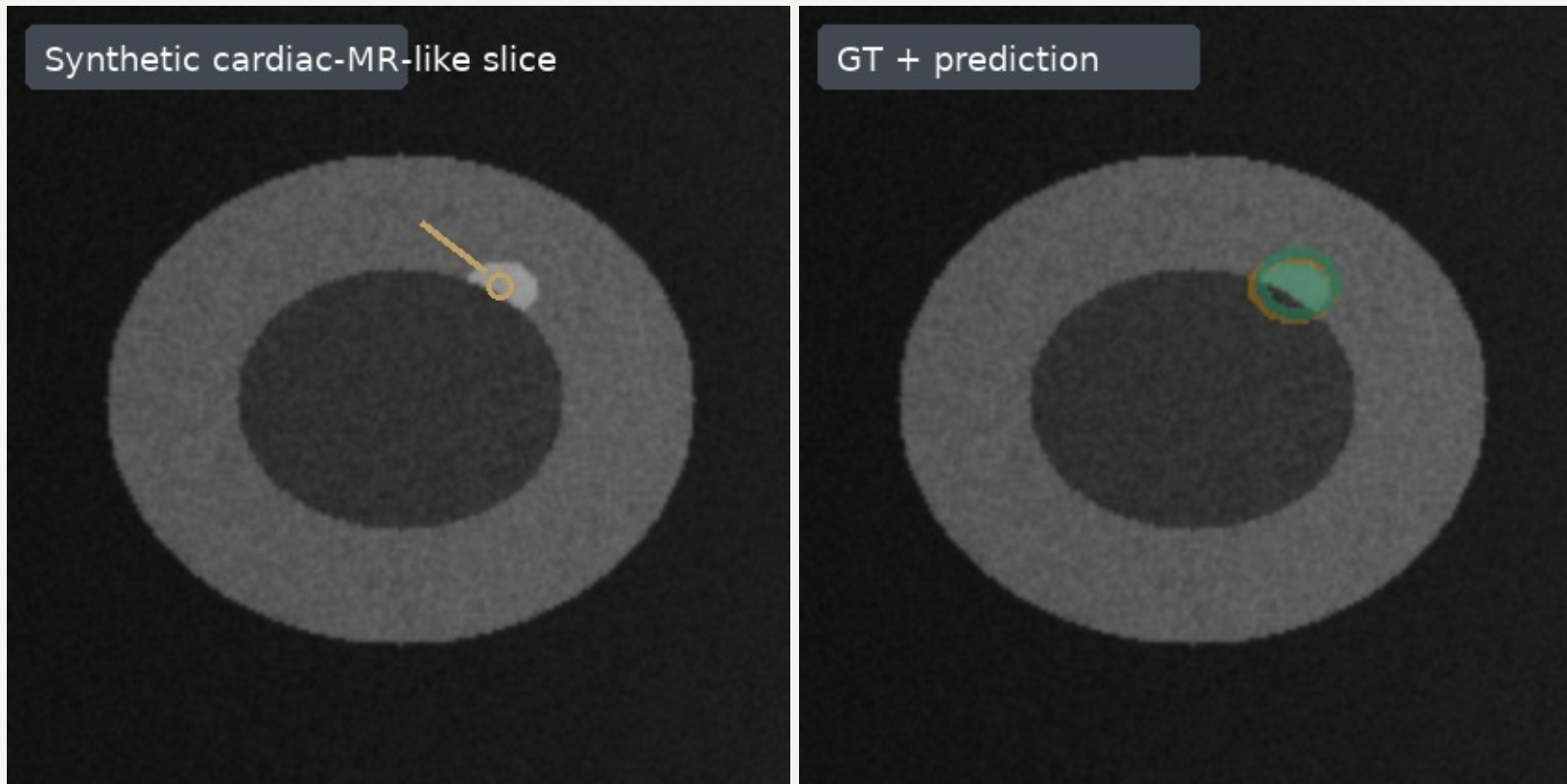


A small lesion is the unit of failure

01/05

Synthetic cardiac-MR-like segmentation task; not clinical validation



Evaluation unit: lesion detection + mask overlap

Dice summarizes mask overlap, but a missed small lesion is counted at the lesion level.

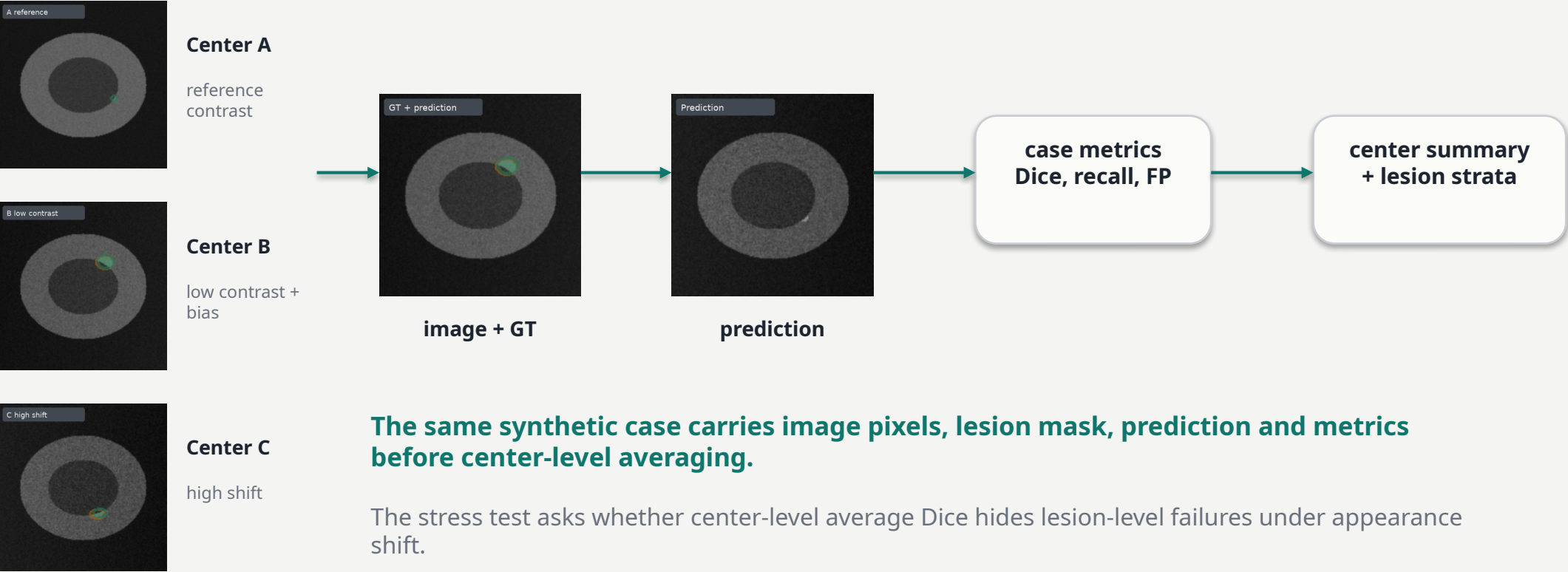
Endpoints: Dice overlap, lesion recall, false-positive burden

Synthetic cardiac-MR-like phantoms; not clinical validation.

■ GT ■ Pred / TP ■ FP ■ FN

The stress test follows one image-to-endpoint path

Center appearance shift propagates through prediction before aggregation

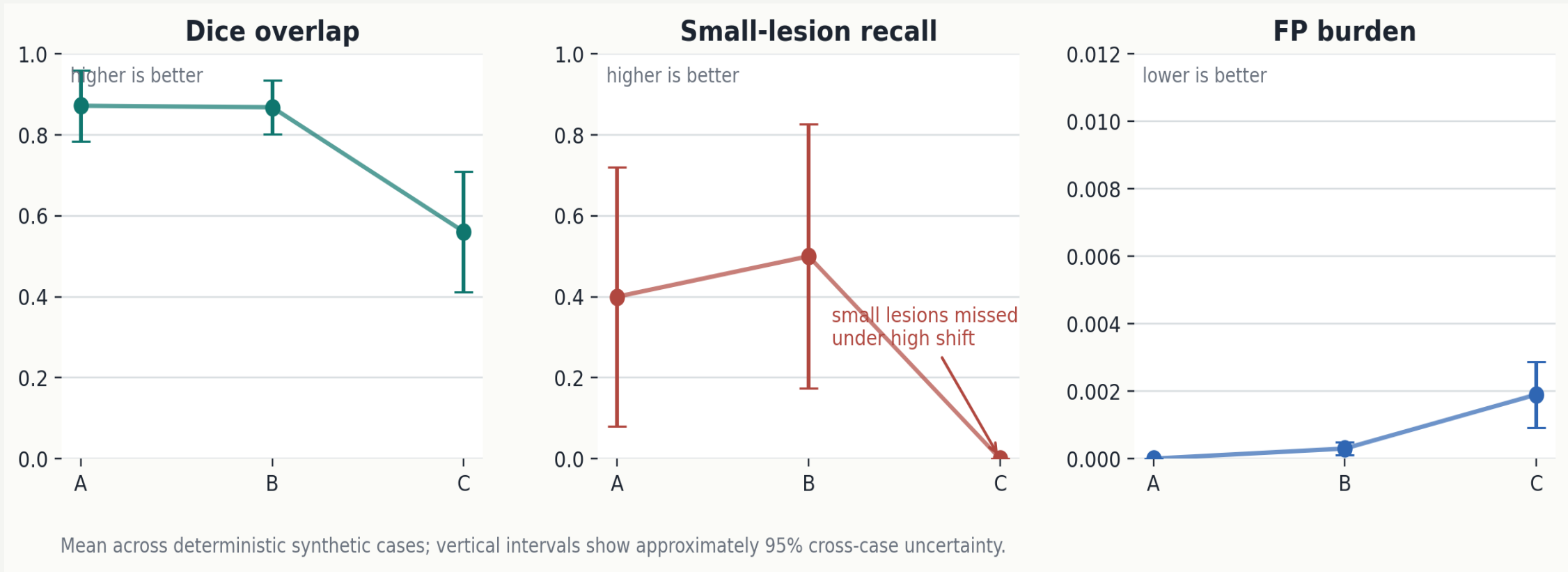


appearance shift

Aggregation is descriptive: no clinical validation claim is made from these phantoms.

Average Dice can hide a lesion-level failure

Endpoint disagreement appears under the high-shift center



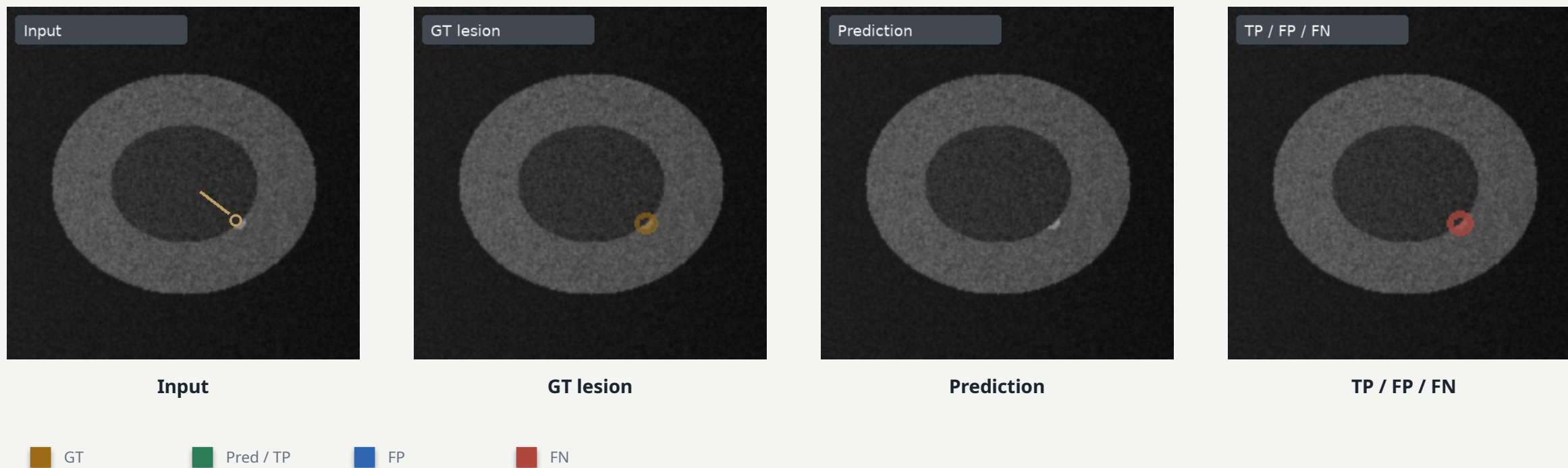
Center C average Dice = 0.56, but small-lesion recall drops to 0.00.

FP burden is lower-is-better; Dice and recall are higher-is-better.

The failure is visible in the same-case overlay

04/05

A missed small lesion drives the negative endpoint



Case C3-03 (Center C, small lesion): Dice 0.00; lesion recall 0; FN pixels 20.

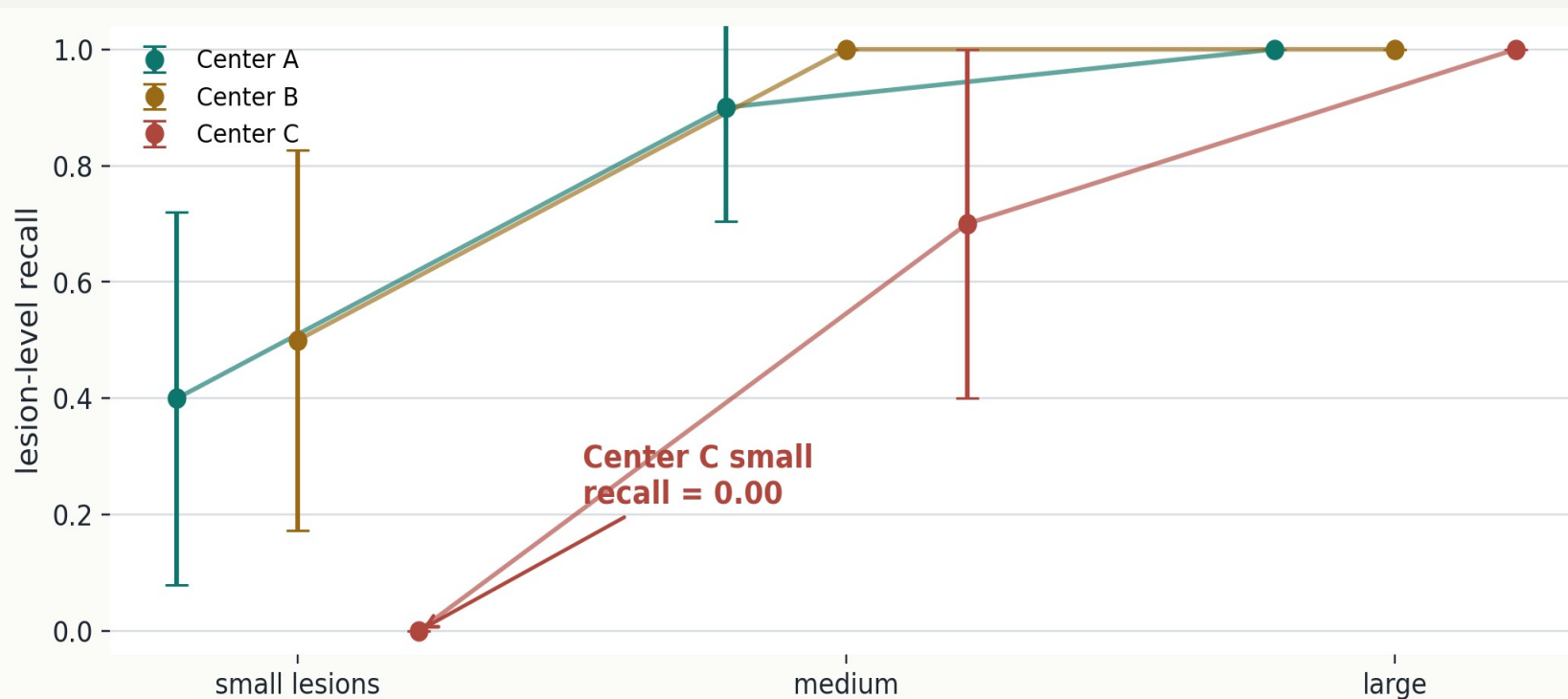
All four panels are the same synthetic slice geometry.

The red FN region shows why an average overlap metric is not enough for this stress condition.

Validate the small-lesion shift explicitly next

05/05

Completed synthetic evidence motivates a planned held-out-center experiment



Completed evidence: deterministic synthetic phantoms only; next validation is planned, not completed.

Completed evidence

Small lesions in the high-shift center are the supported failure regime in this deterministic phantom benchmark.

Planned validation

Hold out center C, stratify by lesion size, and sweep prediction threshold/calibration before comparing methods.

Synthetic cardiac-MR-like phantoms; not a patient validation endpoint.